

Math 221 Homework 11

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Question 6.12. Confirm that the Krylov subspace $K_k(A, y_1)$ has dimension k if and only if the Arnoldi algorithm (Algorithm 6.9) or the Lanczos algorithm (Algorithm 6.10) can compute q_k without quitting first.

Proof. The arguments for Arnoldi and Lanczos are essentially the same, so it is enough to treat Arnoldi. At step j , Arnoldi computes

$$z_j = Aq_j$$

and then orthogonalizes z_j against $q_1, \dots, q_j$ to obtain q_{j+1} , unless the normalization scalar $h_{j+1,j} = \|z_j - \sum_{i=1}^j h_{ij}q_i\|_2$ is zero, in which case the algorithm stops.

Assume first that Arnoldi has not broken down up to step k , so $h_{j+1,j} \neq 0$ for $j = 1, \dots, k-1$. We prove by induction that $\{q_1, \dots, q_j\}$ is an orthonormal basis of $K_j(A, y_1)$ for each $j \leq k$. For $j = 1$, we have $q_1 = y_1/\|y_1\|_2$, so $\text{span}\{q_1\} = K_1(A, y_1)$. Assume $\text{span}\{q_1, \dots, q_j\} = K_j(A, y_1)$. Then $q_j \in K_j(A, y_1)$ implies $z_j = Aq_j \in K_{j+1}(A, y_1)$, and q_{j+1} is a nonzero linear combination of z_j and $q_1, \dots, q_j$, hence also lies in $K_{j+1}(A, y_1)$. Since q_{j+1} is orthogonal to $q_1, \dots, q_j$, it enlarges the basis by one dimension, so

$$\text{span}\{q_1, \dots, q_{j+1}\} = K_{j+1}(A, y_1).$$

Thus $\dim K_k(A, y_1) = k$.

Conversely, suppose $\dim K_k(A, y_1) < k$. Then for some $j < k$ we must have $K_{j+1}(A, y_1) = K_j(A, y_1)$, which means $Aq_j \in \text{span}\{q_1, \dots, q_j\}$. In that case the orthogonalization step produces $z_j - \sum_{i=1}^j h_{ij}q_i = 0$, so $h_{j+1,j} = 0$ and the algorithm stops before computing q_k .

Therefore, $K_k(A, y_1)$ has dimension k if and only if Arnoldi (and similarly Lanczos) can produce q_k without earlier breakdown. $\square$

Question 6.13. Confirm that when $A \in \mathbb{R}^{n \times n}$ is symmetric positive definite and $Q \in \mathbb{R}^{n \times k}$ has full column rank, then

$$T = Q^T A Q$$

is also symmetric positive definite.

Proof. First, T is symmetric because

$$T^T = (Q^T A Q)^T = Q^T A^T Q = Q^T A Q = T,$$

using the symmetry of A .

Now let $x \in \mathbb{R}^k$. Then

$$x^T T x = x^T Q^T A Q x = (Qx)^T A (Qx).$$

Since A is symmetric positive definite, we have $(Qx)^T A (Qx) > 0$ whenever $Qx \neq 0$. If $x \neq 0$, the full column rank of Q implies $Qx \neq 0$, so

$$x^T T x = (Qx)^T A (Qx) > 0.$$

Thus T is symmetric and positive definite. □

Question 6.14. Prove Theorem 6.9. *Remark.* The theorem should use

$$\hat{A} = M^{-1/2}AM^{-1/2},$$

not $M^{-1/2}AM^{1/2}$.

Proof. Let

$$\hat{A} = M^{-1/2}AM^{-1/2}, \quad \hat{b} = M^{-1/2}b,$$

and define the hatted variables by

$$\hat{x}_k = M^{1/2}x_k, \quad \hat{r}_k = M^{-1/2}r_k, \quad \hat{p}_k = M^{1/2}p_k, \quad \hat{z}_k = M^{-1/2}z_k.$$

We show by induction that these are exactly the quantities generated by applying (unpre-conditioned) CG to the system $\hat{A}\hat{x} = \hat{b}$, and that the scalar coefficients ν_k, μ_k are unchanged.

For $k = 1$, we have

$$\hat{z}_1 = \hat{A}\hat{p}_1 = M^{-1/2}AM^{-1/2}M^{1/2}p_1 = M^{-1/2}Ap_1 = M^{-1/2}z_1,$$

so $\hat{z}_1$ and z_1 are related as claimed. The first step size satisfies

$$\hat{\nu}_1 = \frac{\hat{r}_0^\top \hat{r}_0}{\hat{p}_1^\top \hat{z}_1} = \frac{b^\top M^{-1}b}{b^\top M^{-1}z_1} = \nu_1.$$

Thus

$$\hat{x}_1 = \hat{x}_0 + \hat{\nu}_1 \hat{p}_1 = 0 + \nu_1 M^{1/2}p_1 = M^{1/2}x_1,$$

and

$$\hat{r}_1 = \hat{r}_0 - \hat{\nu}_1 \hat{z}_1 = M^{-1/2}r_0 - \nu_1 M^{-1/2}z_1 = M^{-1/2}r_1.$$

The next coefficient is

$$\hat{\mu}_2 = \frac{\hat{r}_1^\top \hat{r}_1}{\hat{r}_0^\top \hat{r}_0} = \frac{r_1^\top M^{-1}r_1}{r_0^\top M^{-1}r_0} = \mu_2,$$

so

$$\hat{p}_2 = \hat{r}_1 + \hat{\mu}_2 \hat{p}_1 = M^{-1/2}r_1 + \mu_2 M^{1/2}p_1 = M^{1/2}p_2.$$

This establishes the relations for the first step.

Assume now that for some $k \geq 1$ the identities

$$\hat{x}_j = M^{1/2}x_j, \quad \hat{r}_j = M^{-1/2}r_j, \quad \hat{p}_j = M^{1/2}p_j$$

hold for all $j \leq k$. Then

$$\hat{z}_k = \hat{A}\hat{p}_k = M^{-1/2}AM^{-1/2}M^{1/2}p_k = M^{-1/2}z_k.$$

The step size at iteration k is

$$\hat{\nu}_k = \frac{\hat{r}_{k-1}^\top \hat{r}_{k-1}}{\hat{p}_k^\top \hat{z}_k} = \frac{r_{k-1}^\top M^{-1}r_{k-1}}{p_k^\top z_k} = \nu_k,$$

so

$$\widehat{x}_k = \widehat{x}_{k-1} + \widehat{\nu}_k \widehat{p}_k = M^{1/2} x_{k-1} + \nu_k M^{1/2} p_k = M^{1/2} x_k,$$

and

$$\widehat{r}_k = \widehat{r}_{k-1} - \widehat{\nu}_k \widehat{z}_k = M^{-1/2} r_{k-1} - \nu_k M^{-1/2} z_k = M^{-1/2} r_k.$$

Similarly,

$$\widehat{\mu}_{k+1} = \frac{\widehat{r}_k^\top \widehat{r}_k}{\widehat{r}_{k-1}^\top \widehat{r}_{k-1}} = \frac{r_k^\top M^{-1} r_k}{r_{k-1}^\top M^{-1} r_{k-1}} = \mu_{k+1},$$

and hence

$$\widehat{p}_{k+1} = \widehat{r}_k + \widehat{\mu}_{k+1} \widehat{p}_k = M^{-1/2} r_k + \mu_{k+1} M^{1/2} p_k = M^{1/2} p_{k+1}.$$

Thus the relations hold at step $k+1$, completing the induction. Therefore the preconditioned CG iterates correspond exactly to the CG iterates for $\widehat{A}\widehat{x} = \widehat{b}$ under the change of variables above, which is the content of Theorem 6.9. $\square$

Question 6.18. Fill in a table like Table 6.1 for Poisson's equation in three dimensions on an $N \times N \times N$ grid, so $n = N^3$.

A suitable 3-D analogue of Table 6.1 is

Method	Serial time	Space
Dense Cholesky	n^3	n^2
Explicit inverse	n^2	n^2
Band Cholesky	$n^{7/3}$	$n^{5/3}$
Jacobi	$n^{5/3}$	n
Gauss–Seidel	$n^{5/3}$	n
Sparse Cholesky	n^2	$n^{4/3}$
Conjugate gradients	$n^{4/3}$	n
Successive overrelaxation	$n^{4/3}$	n
SSOR with Chebyshev accel.	$n^{7/6}$	n
Fast Fourier transform	$n \log n$	n
Block cyclic reduction	$n \log n$	n
Multigrid	n	n
Lower bound	n	n

Table 1. Asymptotic cost for Poisson on an $N \times N \times N$ grid, $n = N^3$.

Proof. Write $n = N^3$.

Dense methods. Dense Cholesky and explicit inversion ignore sparsity, so the $n \times n$ system has cost $O(n^3)$ flops and $O(n^2)$ storage; once an explicit inverse is available, each solve is a dense matrix–vector multiply, $O(n^2)$ time and $O(n^2)$ space.

Banded Cholesky. With lexicographic ordering the 3-D Laplacian has half-bandwidth $m \sim N^2$ (neighbors in the third direction are N^2 indices apart). Cholesky of an $n \times n$ SPD band matrix with bandwidth m costs $O(nm^2)$ flops and $O(nm)$ storage:

$$nm^2 \sim N^3(N^2)^2 = N^7 = n^{7/3}, \quad nm \sim N^3N^2 = N^5 = n^{5/3}.$$

Jacobi / Gauss–Seidel. For the d -dimensional Poisson problem the spectral radius of these stationary iterations satisfies $1 - \rho \sim N^{-2}$, so one needs $O(N^2)$ iterations. Each sweep touches all $n = N^3$ unknowns, giving time $O(nN^2) = O(N^5) = O(n^{5/3})$ and $O(n)$ storage.

Sparse Cholesky. With nested dissection on a 3-D grid, separators are $O(N^2)$ -sized surfaces. The factorization cost is $O(N^6)$ flops and $O(N^4)$ storage, i.e.

$$O(N^6) = O((N^3)^2) = O(n^2), \quad O(N^4) = O((N^3)^{4/3}) = O(n^{4/3}).$$

CG and SOR. For the discrete Poisson matrix, the condition number satisfies $\kappa(A) \sim h^{-2} \sim N^2$, independent of dimension. Conjugate gradients and optimally tuned SOR both need $O(\sqrt{\kappa}) = O(N)$ iterations, each of $O(n)$ cost, so the time is $O(nN) = O(N^4) = O(n^{4/3})$ with $O(n)$ storage.

SSOR + Chebyshev. Chebyshev acceleration reduces the iteration count to $O(\kappa^{1/4}) \sim O(N^{1/2})$, so the total time is $O(nN^{1/2}) = O(N^{3.5}) = O(n^{7/6})$, again with $O(n)$ storage.

FFT and block cyclic reduction. The discrete sine/FFT-based solvers in 3-D require $O(N^3 \log N)$ work and $O(N^3)$ storage, that is $O(n \log n)$ time and $O(n)$ memory. Block cyclic reduction has the same order.

Multigrid and lower bound. A multigrid V-cycle visits $O(n)$ unknowns at each level, and the number of cycles needed for a fixed accuracy is bounded independently of N , so the overall work and storage are both $O(n)$. Any method must at least read and store all n unknowns, so the lower bound is also $O(n)$ in both time and space. $\square$